

```
#Load required libraries
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(ggplot2)
library(knitr)
setwd("/Users/aka6208/Desktop/game assignment")
```

```
#Question 1
# Read CSV in the file
players <- read.csv("players.csv")
sessions <- read.csv("sessions.csv")
games <- read.csv("games.csv")
#Then we need to check the data sets
head(players)
```

```
##   player_id age   country signup_date
## 1         1  54     USA    2023-06-26
## 2         2  44     India  2023-08-08
## 3         3  30     India  2022-09-19
## 4         4  58     UK     2022-03-30
## 5         5  23 Australia 2024-07-03
## 6         6  36 Australia 2022-07-27
```

```
head(sessions)
```

```
##   session_id player_id game_id play_time_minutes score
## 1         1         1    3585          11           26   664
## 2         2         2     348          27           73   319
## 3         3         3    1264           4           73   603
## 4         4         4    4666          23           67   554
## 5         5         5    2743          23           79   781
## 6         6         6    2074          49          111   481
```

```
str(players)
```

```
## 'data.frame':   5000 obs. of  4 variables:
##  $ player_id  : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ age        : int  54 44 30 58 23 36 54 34 38 26 ...
##  $ country     : chr  "USA" "India" "India" "UK" ...
##  $ signup_date: chr  "2023-06-26" "2023-08-08" "2022-09-19" "2022-03-30" ...
```

```
str(sessions)
```

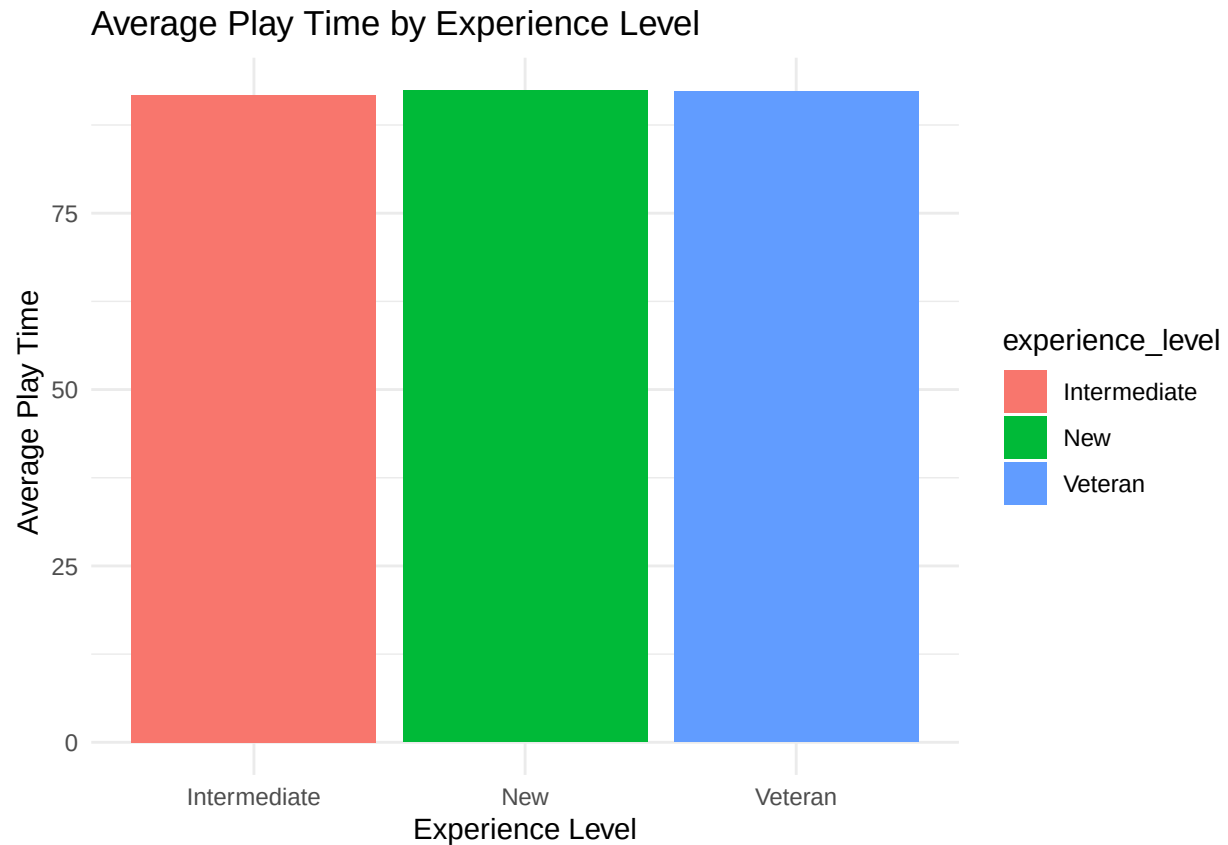
```
## 'data.frame': 20000 obs. of 5 variables:
## $ session_id : int 1 2 3 4 5 6 7 8 9 10 ...
## $ player_id : int 3585 348 1264 4666 2743 2074 3149 1352 2822 3828 ...
## $ game_id : int 11 27 4 23 23 49 34 21 22 1 ...
## $ play_time_minutes: int 26 73 73 67 79 111 23 106 130 131 ...
## $ score : int 664 319 603 554 781 481 247 943 67 269 ...
```

```
#The signup_date needs to be converted into Date format
players$signup_date <- as.Date(players$signup_date)
#
players <- players %>%
mutate(
  experience_level = case_when(
    format(signup_date, "%Y") < "2023" ~ "Veteran",
    format(signup_date, "%Y") == "2023" ~ "Intermediate",
    format(signup_date, "%Y") > "2023" ~ "New"
  )
)
merged_data <- sessions %>%
  left_join(players, by = "player_id")

summary_table <- merged_data %>%
  group_by(experience_level) %>%
  summarise(
    number_players = n_distinct(player_id),
    average_play_time = mean(play_time_minutes, na.rm = TRUE),
    average_score = mean(score, na.rm = TRUE)
  )
kable(summary_table)
```

experience_level	number_players	average_play_time	average_score
Intermediate	1825	91.72935	498.2044
New	1340	92.41463	495.9630
Veteran	1764	92.26217	501.0407

```
ggplot(summary_table,
  aes(
    x = experience_level,
    y = average_play_time,
    fill = experience_level
  )) +
  geom_bar(stat = "identity") +
  labs(
    title = "Average Play Time by Experience Level",
    x = "Experience Level",
    y = "Average Play Time"
  ) +
  theme_minimal()
```



#Question 2

```
# Take the sessions dataframe
# then join the games dataframe
# using the common column game_id
genre_data <- sessions %>%
  left_join (games, by = "game_id")

#take the joined data
#then group bu the data by genre
genre_summary <- genre_data %>%
  group_by (genre) %>%

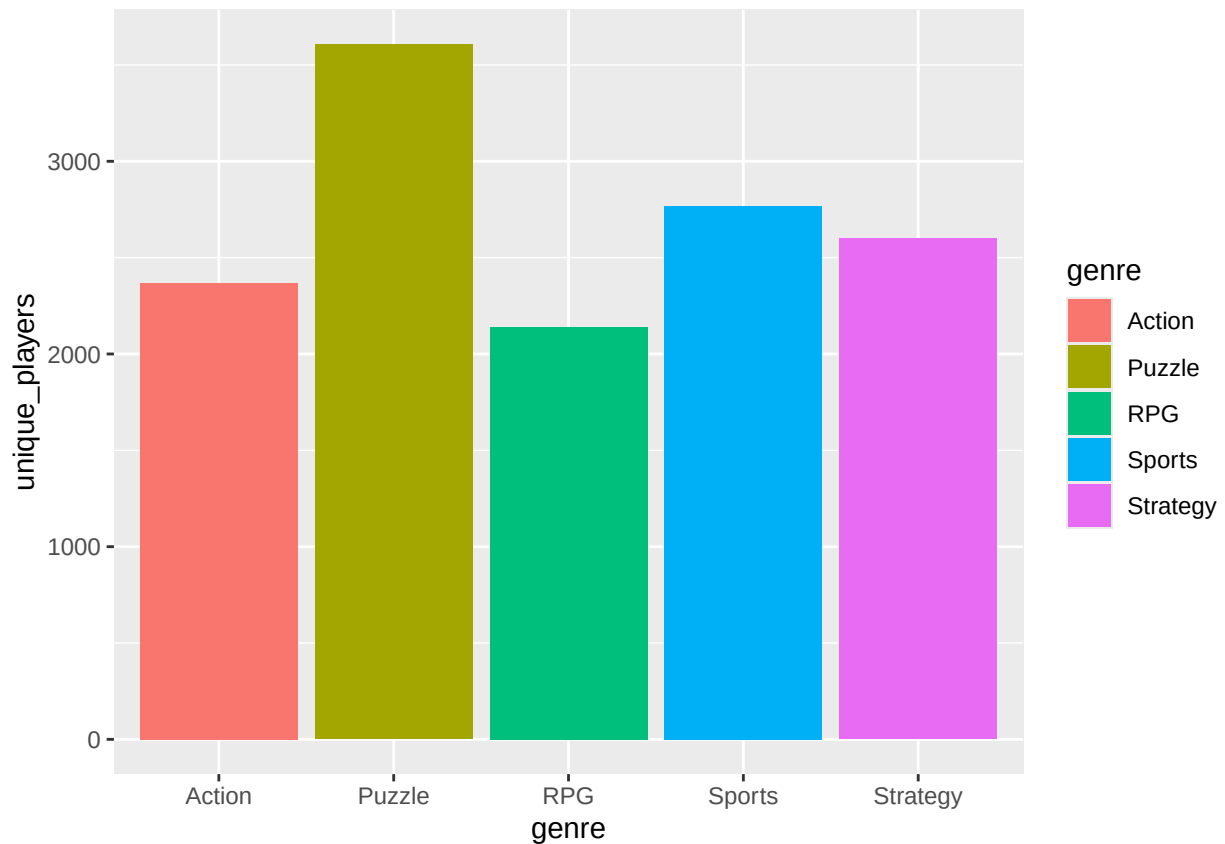
summarise(

  average_play_time = mean(play_time_minutes, na.rm = TRUE),
  average_score = mean(score, na.rm= TRUE),
  number_session = n(),
  unique_players = n_distinct(player_id)
)
kable(genre_summary)
```

genre	average_play_time	average_score	number_session	unique_players
Action	92.09816	493.0569	3199	2370

genre	average_play_time	average_score	number_session	unique_players
Puzzle	91.67226	496.5239	6371	3606
RPG	91.41153	499.6938	2792	2139
Sports	92.58573	503.3790	3995	2769
Strategy	92.88334	500.8435	3643	2599

```
ggplot(
  genre_summary,
  aes(
    x=genre,
    y=unique_players,
    fill = genre
  )
)+
  geom_bar(stat = "identity"
)
```



#Question 3

```
top_players <- sessions %>%
  # Take the sessions dataframe
  group_by(player_id) %>%
  #Then group the data by player_id
```

```

summarise(
  total_play_time = sum(play_time_minutes, na.rm = TRUE)
) %>%
#create a summary table and calculate total play time for each player

arrange(desc(total_play_time)) %>%
# arrange players from highest to lowest play time

slice_head (n=10)
# Select the top 10 players only

top_player_data <- sessions %>%
#Take the sessions dataframe

filter(player_id %in% top_players$player_id)

top_summary <- top_player_data %>%
group_by(player_id) %>%
#create summary statistics

summarise(
total_sessions = n(),

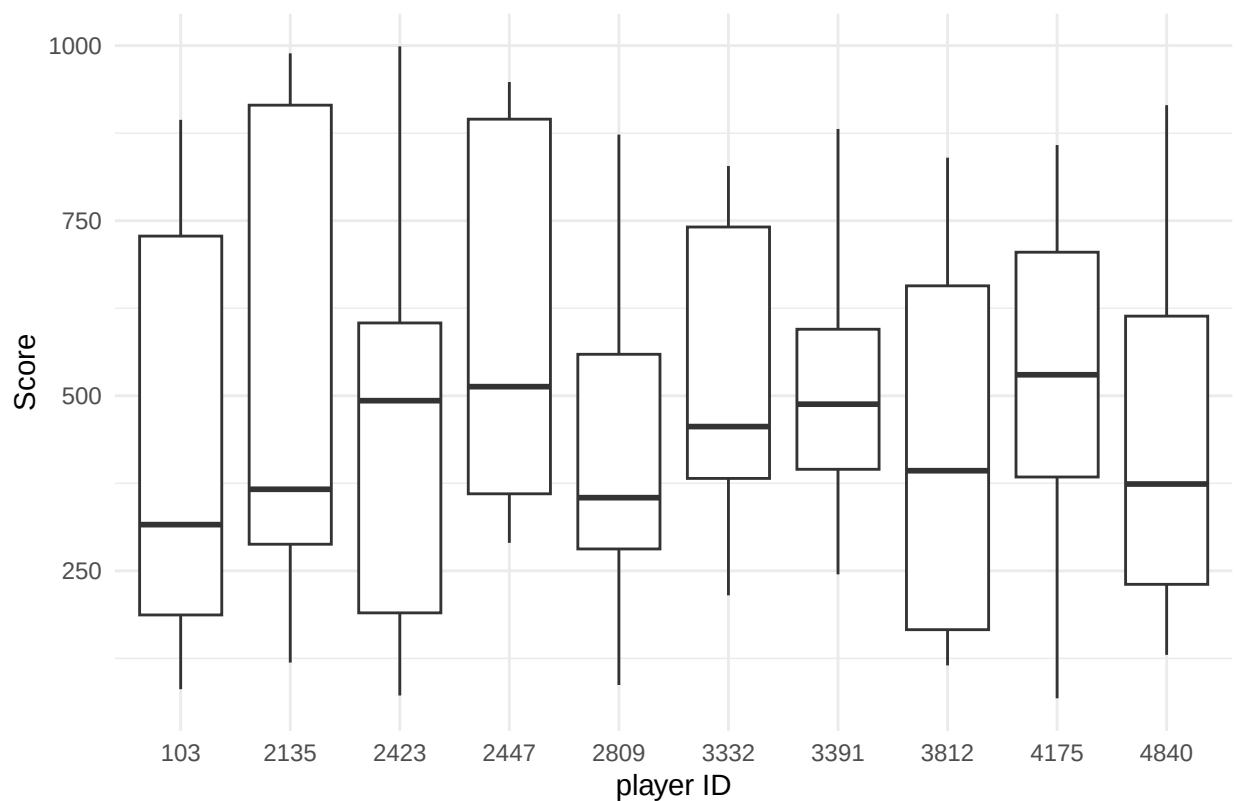
average_play_time = mean(play_time_minutes, na.rm = TRUE),

average_score = mean (score, na.rm=TRUE)
)
ggplot(
top_player_data,
aes(
x=factor(player_id),
y=score
)
) +
geom_boxplot()+
labs(
title = "Score Distribution of Top 10 Players",
  x="player ID",
  y="Score"
) +

theme_minimal()

```

Score Distribution of Top 10 Players



```
#Question 4
players <- players %>%
  mutate(
    age_group = case_when(
      age >= 16 & age <= 25 ~ "16-25",
      age >= 26 & age <= 35 ~ "26-35",
      age >= 36 & age <= 45 ~ "36-45",
      age >= 46 ~ "46+"
    )
  )

age_data <- sessions %>%
  left_join(players, by="player_id")
#join sessions data with players data
age_summary <- age_data %>%
  group_by(age_group) %>%
  summarise(
    average_play_time=mean(play_time_minutes, na.rm = TRUE),
    average_score=mean(score, na.rm = TRUE),
    number_sessions = n()
  )

kable(age_summary)
```

age_group	average_play_time	average_score	number_sessions
16-25	91.79946	496.6914	4413
26-35	91.56831	503.5342	4538
36-45	91.57548	498.0227	4756
46+	93.11298	496.7149	6293

```
ggplot(age_summary,
  aes(
    x=age_group,
    y=average_play_time,
    fill = age_group
  )
)+
geom_bar(stat = "identity") +
labs(
  title= "Average Play Time by Age Group",
  x="Age Group",
  y="Average Play Time"
)+
theme_minimal()
```

